Duration Models: From Hazards to Treatment Effects

Motivated by Abbring & Van den Berg (2003, *Econometrica*)

Econometrics Reading Group

Why duration models?

- Many economic outcomes are about **how long until something happens**:
  - unemployment → job; poverty → exit; firm → closure; marriage → divorce.
- The questions are about the **rate** of leaving:
  - how it changes with time already spent;
  - how it responds to events along the way.

Destination of this talk: Abbring & Van den Berg (2003)

- The effect of a treatment that *arrives during the spell* (e.g. a benefits reduction) on the exit rate.
- To get there, we build the objects one at a time.

Road map: the questions, in order

1. **How do we describe a duration?**
   Hazard and survival. Cox (1972), Lancaster (1979)

2. **People differ: is a falling exit rate a trap or sorting?**
   Unobserved heterogeneity, dynamic selection. Lancaster (1979), Heckman & Singer (1984)

3. **When can the two be separated?**
   The mixed proportional hazard model. Elbers & Ridder (1982)

4. **Two durations racing.**
   Competing risks. Cox (1959), Tsiatis (1975), Heckman & Honoré (1989)

5. **A treatment arriving during the spell.**
   Abbring & Van den Berg (2003)

One duration: hazard and survival

$Y \geq 0$ continuous, $S(y) := \Pr(Y > y)$ survival, f density.

$$\theta(y) := \lim_{dy \downarrow 0} \frac{\Pr(y \leq Y < y + dy \mid Y \geq y)}{dy} = \frac{f(y)}{S(y)} \qquad \text{[conditional probability]}$$

$$= -\frac{d}{dy} \log S(y) \qquad [S' = -f]$$

$$\Theta(y) := \int_0^y \theta(u)\, du = -\log S(y) \qquad \text{[integrate; } S(0) = 1\text{]}$$

$$\implies \quad S(y) = e^{-\Theta(y)}$$

- $\theta(y)$: exit rate *among those still unemployed*. The effect will act on θ.
- Constant hazard $\theta = c$: $S(y) = e^{-cy}$, memoryless.
- θ decreasing: **negative duration dependence**.

People differ: individual vs. population hazard

Unobserved type $V \geq 0$, $V \sim G$. Assume

$$\theta(y \mid V) = \lambda(y)\, V, \qquad \lambda(y) = \theta(y \mid V = 1), \qquad S(y \mid V) = e^{-\Lambda(y) V}.$$

What we observe is the **population**:

$$\bar{S}(y) = \mathbb{E}\big[S(y \mid V)\big] = \mathbb{E}\big[e^{-\Lambda(y)V}\big] \qquad \text{[LIE]}$$

$$\bar{\theta}(y) = -\frac{\bar{S}'(y)}{\bar{S}(y)} = \lambda(y)\, \frac{\mathbb{E}\big[V e^{-\Lambda(y)V}\big]}{\mathbb{E}\big[e^{-\Lambda(y)V}\big]} \qquad \text{[differentiate under } \mathbb{E}\text{]}$$

- The ratio is an average of V, weighting each type by its survival $e^{-\Lambda(y)V}$.
- Next: it is the average type **among those still unemployed**.

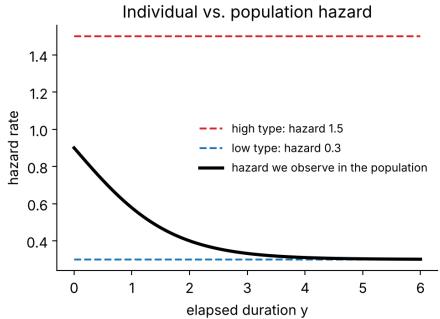
Who is still unemployed?

$$\Pr(V = v \mid Y > y) = \frac{S(y \mid v)\Pr(V = v)}{\bar{S}(y)} \qquad \text{[Bayes]}$$

$$\mathbb{E}[V \mid Y > y] = \frac{\mathbb{E}[Ve^{-\Lambda(y)V}]}{\mathbb{E}[e^{-\Lambda(y)V}]} \qquad \text{[its mean]}$$

$$\implies \bar{\theta}(y) = \lambda(y)\,\mathbb{E}[V \mid Y > y] \qquad \text{[previous slide]}$$

- Prior $\Pr(V = v) \times$ likelihood $S(y \mid v)$: staying is evidence of a low v.
- Average individual hazard: $\lambda(y)\mathbb{E}[V]$.
- Observed hazard: $\lambda(y)\mathbb{E}[V \mid Y > y]$.
- Gap = **dynamic selection**.



Two types, constant hazards. Nobody's hazard falls; the observed one does.

The Laplace transform: isolating the type distribution

$$\mathcal{L}(a) := \mathbb{E}\big[e^{-aV}\big], \qquad \bar{S}(y) = \mathcal{L}\big(\Lambda(y)\big).$$

- $\mathcal{L}$ depends only on G; Λ only on duration dependence.
- Survival = (heterogeneity) applied to (duration dependence).

Properties

$$-\frac{\mathcal{L}'(a)}{\mathcal{L}(a)} = \mathbb{E}[V \mid \text{survived to } a] \qquad \text{[differentiate under } \mathbb{E}\text{]}$$

$$\frac{d^2}{da^2}\log \mathcal{L}(a) = \mathrm{Var}(V \mid \text{survived to } a) \qquad \text{[quotient rule]}$$

$$\implies \frac{d}{dy}\mathbb{E}[V \mid Y > y] = -\lambda(y)\,\mathrm{Var}(V \mid Y > y) \;\le 0 \qquad \text{[chain rule]}$$

- **Uniqueness:** $\mathcal{L}$ determines G.
- **Analyticity:** $\mathcal{L}$ on any small interval $\Rightarrow$ $\mathcal{L}$ everywhere.
- Strategy: learn the index, learn $\mathcal{L}$ on an interval $\Rightarrow$ G.

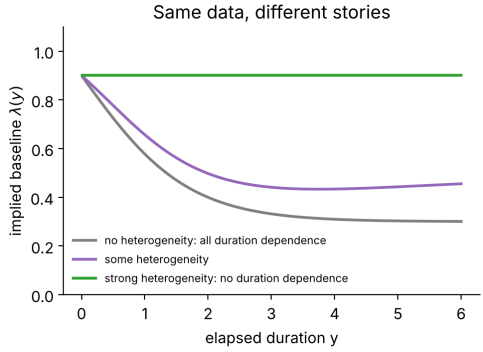
Trap or sorting? Not identified from one duration

$$\bar{\theta}(y) = \underbrace{\lambda(y)}_{\text{duration dep.}} \cdot \underbrace{\mathbb{E}[V \mid Y > y]}_{\text{sorting}}$$

- Data: $\bar{S}(y)$ for all y.
- Unknowns: Λ and G, with $\bar{S}(y) = \mathcal{L}_G(\Lambda(y))$.
- One data function, two unknown functions of the same y.
- Any G fits: Λ bends to absorb the rest.

To separate them, we need variation that moves one but not the other:

- groups at different stages of sorting, same λ;
- the same person observed twice;
- parametric restrictions on Λ or G.



Same observed survival, three type distributions, three implied baselines.